ANOVA Case Studies Tutorial

Lee Barnby

2025-10-30

Introduction

This tutorial covers various case studies involving one-way Analysis of Variance (ANOVA) using R. We will go through setting up the data, performing ANOVA, and interpreting the results.

Case Study 1: Insect Spray Data

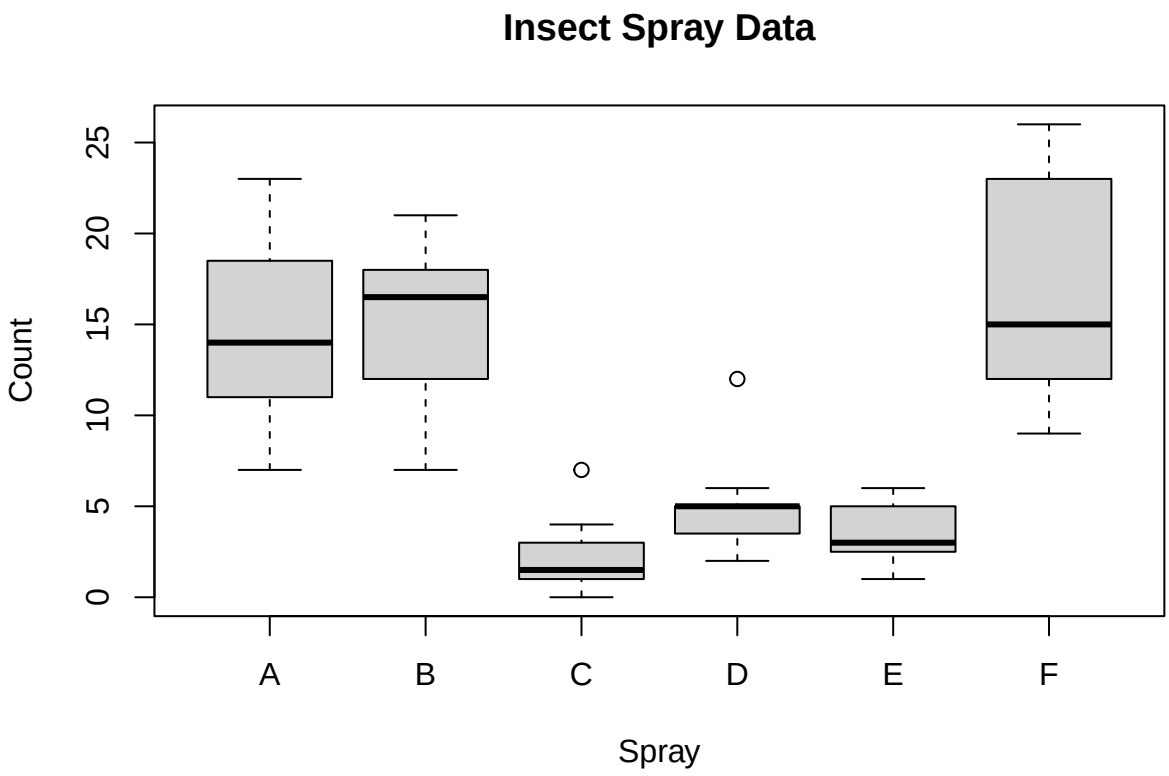
The insect spray data is a built-in dataset in R, so we will use it directly without detailing the data setup.

Steps

1. **Box and whisker plot**
2. **Null hypothesis and oneway.test**
3. **Bonferroni p-value adjustment**
4. **Bartlett K-squared test of homogeneity of variances**
5. **Non-parametric alternative: Kruskal-Wallis one-way ANOVA**
6. **Multiple comparisons: pairwise.t.test and Tukey HSD**
7. **Relationship with linear regression**
8. **Assessing normal distribution**

```r
# Load the dataset
data(InsectSprays)

# Box and whisker plot
boxplot(count ~ spray, data = InsectSprays, main = "Insect Spray Data", xlab = "Spray", ylab = "Count")
```




```r
# ANOVA test
anova_result <- aov(count ~ spray, data = InsectSprays)
summary(anova_result)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## spray        5   2669   533.8    34.7 <2e-16 ***
## Residuals   66   1015    15.4
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Bonferroni adjustment
pairwise.t.test(InsectSprays$count, InsectSprays$spray, p.adjust.method = "bonferroni")
```

```
##
##  Pairwise comparisons using t tests with pooled SD
##
## data:  InsectSprays$count and InsectSprays$spray
##
##   A       B       C       D       E
## B 1       -       -       -       -
## C 1.1e-09 1.3e-10 -       -       -
## D 1.5e-06 1.8e-07 1       -       -
## E 4.1e-08 4.9e-09 1       1       -
## F 1       1       4.2e-12 6.1e-09 1.6e-10
##
## P value adjustment method: bonferroni
```

```r
# Bartlett test for homogeneity of variances
bartlett.test(count ~ spray, data = InsectSprays)
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  count by spray
## Bartlett's K-squared = 25.96, df = 5, p-value = 9.085e-05
```

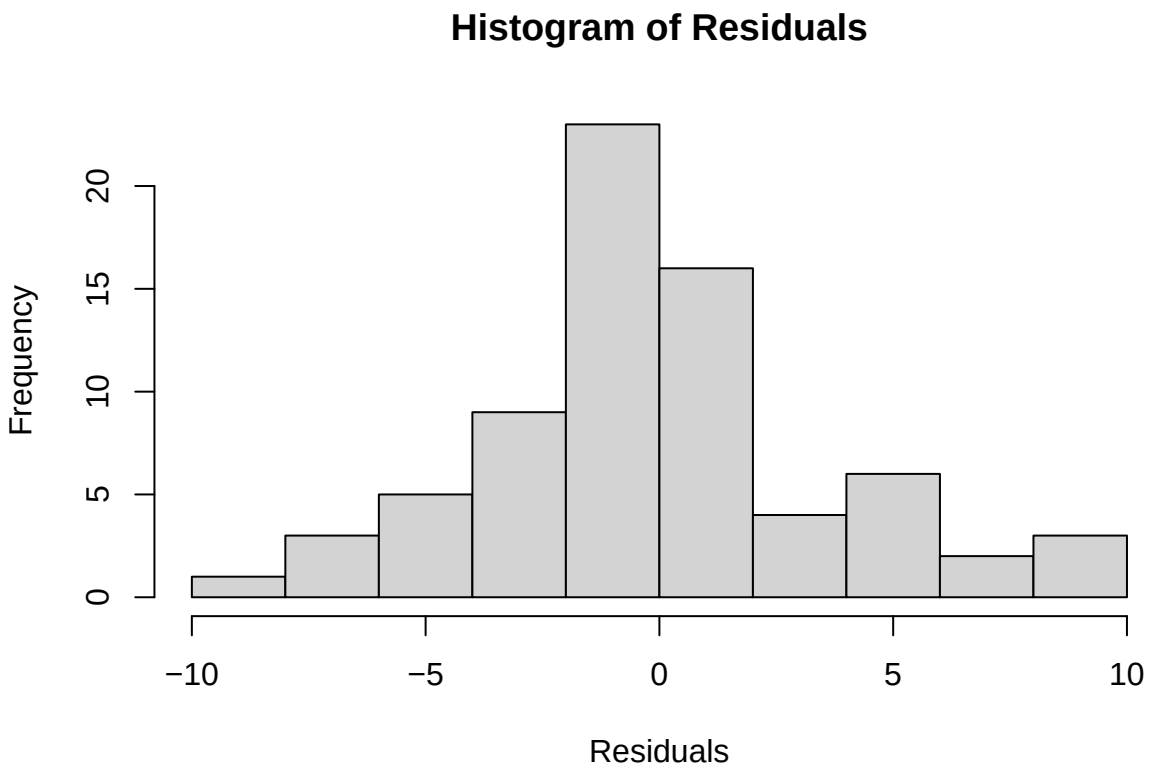
```r
# Kruskal-Wallis test
kruskal.test(count ~ spray, data = InsectSprays)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  count by spray
## Kruskal-Wallis chi-squared = 54.691, df = 5, p-value = 1.511e-10
```

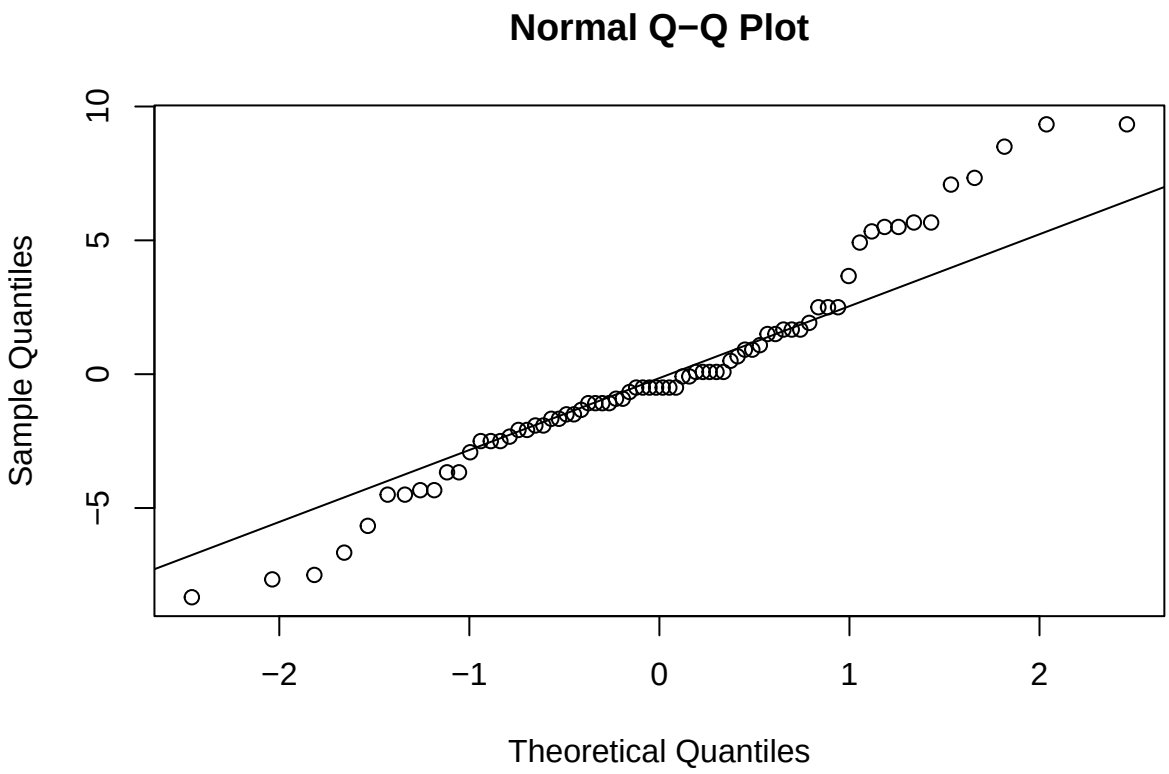
```r
# Tukey HSD test
TukeyHSD(anova_result)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = count ~ spray, data = InsectSprays)
##
## $spray
##            diff        lwr       upr     p adj
## B-A   0.8333333  -3.866075  5.532742 0.9951810
## C-A -12.4166667 -17.116075 -7.717258 0.0000000
## D-A  -9.5833333 -14.282742 -4.883925 0.0000014
## E-A -11.0000000 -15.699409 -6.300591 0.0000000
## F-A   2.1666667  -2.532742  6.866075 0.7542147
## C-B -13.2500000 -17.949409 -8.550591 0.0000000
## D-B -10.4166667 -15.116075 -5.717258 0.0000002
## E-B -11.8333333 -16.532742 -7.133925 0.0000000
## F-B   1.3333333  -3.366075  6.032742 0.9603075
## D-C   2.8333333  -1.866075  7.532742 0.4920707
## E-C   1.4166667  -3.282742  6.116075 0.9488669
## F-C  14.5833333   9.883925 19.282742 0.0000000
## E-D  -1.4166667  -6.116075  3.282742 0.9488669
## F-D  11.7500000   7.050591 16.449409 0.0000000
## F-E  13.1666667   8.467258 17.866075 0.0000000
```

```r
# Assessing normal distribution
hist(residuals(anova_result), main = "Histogram of Residuals", xlab = "Residuals")
```




```
qqnorm(residuals(anova_result))
qqline(residuals(anova_result))
```




```r
shapiro.test(residuals(anova_result))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(anova_result)
## W = 0.96006, p-value = 0.02226
```

Case Study 2: Reporting of Migraine Pain

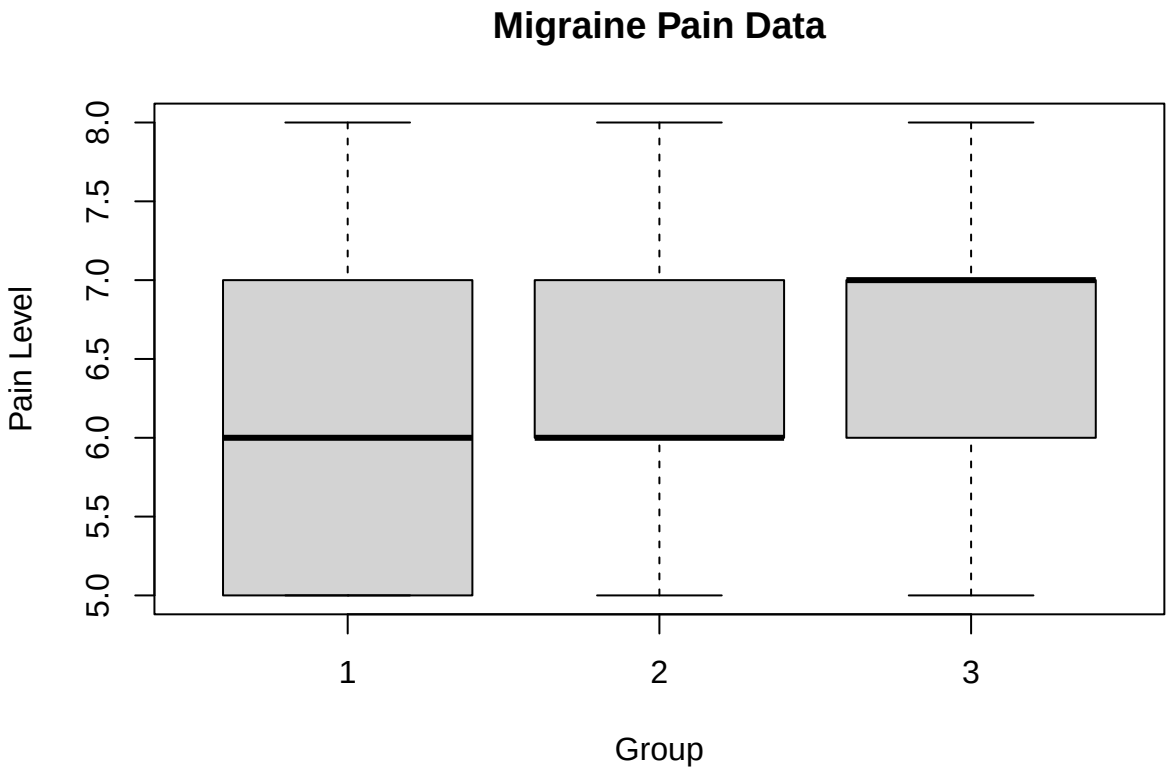
Three sets of 9 volunteers who suffered from headaches reported their migraine pain.

Steps

1. **Setting up the data in R**
2. **Box and whisker plot**
3. **ANOVA test**
4. **Post-hoc tests**

```r
# Example data
migraine_data <- data.frame(
  group = factor(rep(1:3, each = 9)),
  pain = c(5, 6, 7, 8, 5, 6, 7, 8, 5, 6, 7, 8, 5, 6, 7, 8, 5, 6, 7, 8, 5, 6, 7, 8, 5, 6, 7)
)

# Box and whisker plot
boxplot(pain ~ group, data = migraine_data, main = "Migraine Pain Data", xlab = "Group", ylab = "Pain L
```




```
# ANOVA test
anova_result <- aov(pain ~ group, data = migraine_data)
summary(anova_result)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## group        2   0.22  0.1111   0.082  0.921
## Residuals   24  32.44  1.3519
```

```
# Post-hoc tests
TukeyHSD(anova_result)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = pain ~ group, data = migraine_data)
##
## $group
##          diff       lwr      upr     p adj
## 2-1 0.1111111 -1.257647 1.479869 0.9776204
## 3-1 0.2222222 -1.146536 1.590981 0.9136978
## 3-2 0.1111111 -1.257647 1.479869 0.9776204
```

Case Study 3: Length of Stay in Hospitals

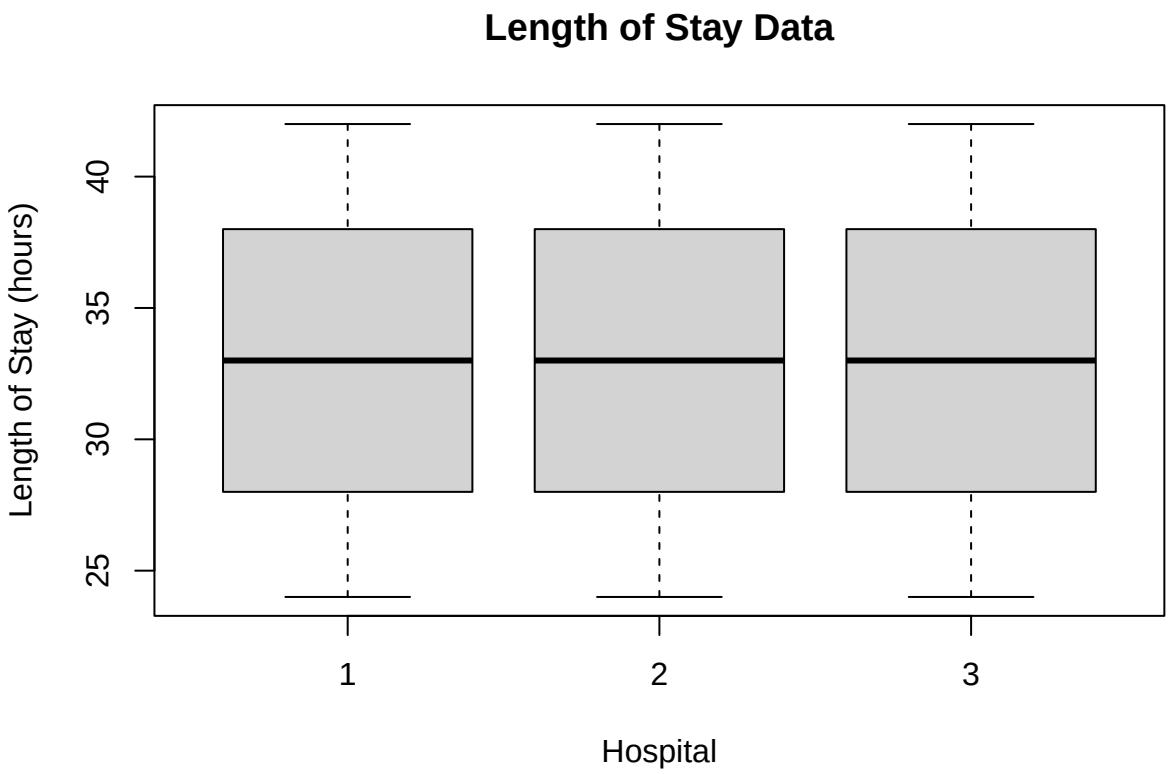
The length of stay (in hours) for patients having had the same surgery in three different hospitals.

Steps

1. **Setting up the data in R**
2. **Box and whisker plot**
3. **ANOVA test**
4. **Post-hoc tests**

```
# Example data
stay_data <- data.frame(
  hospital = factor(rep(1:3, each = 10)),
  stay_length = c(24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 24, 26
)

# Box and whisker plot
boxplot(stay_length ~ hospital, data = stay_data, main = "Length of Stay Data", xlab = "Hospital", ylab
```




```
# ANOVA test
anova_result <- aov(stay_length ~ hospital, data = stay_data)
summary(anova_result)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## hospital     2      0    0.00       0      1
## Residuals   27    990   36.67
```

```
# Post-hoc tests
TukeyHSD(anova_result)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
```

```
##
## Fit: aov(formula = stay_length ~ hospital, data = stay_data)
##
## $hospital
##     diff       lwr      upr p adj
## 2-1    0 -6.714295 6.714295     1
## 3-1    0 -6.714295 6.714295     1
## 3-2    0 -6.714295 6.714295     1
```

Case Study 4: Machine Output

Given the following data relating to the output from 3 machines, can we conclude that average output from the machines are different?

Data

```
machine_data <- data.frame(
  machine = factor(rep(1:3, each = 5)),
  output = c(53, 47, 46, 50, 49, 61, 55, 52, 58, 54, 51, 51, 49, 54, 50)
)

# Box and whisker plot


# ANOVA test

# Post-hoc tests
TukeyHSD(anova_result)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = stay_length ~ hospital, data = stay_data)
##
## $hospital
##     diff       lwr      upr p adj
## 2-1    0 -6.714295 6.714295     1
## 3-1    0 -6.714295 6.714295     1
## 3-2    0 -6.714295 6.714295     1
```

Case Study 5: Teaching Methods

In the teaching of arithmetic, 45 pupils were divided into five groups of nine. Groups A and B were taught as normal. Groups C, D, and E were, respectively, praised publicly, reproved publicly, and ignored.

Data

```
teaching_data <- data.frame(
  group = factor(rep(1:5, each = 9)),
  score = c(17, 14, 24, 20, 24, 23, 16, 15, 24, 21, 23, 13, 19, 13, 19, 20, 21, 16, 28, 30, 29, 24, 27,
)
```

```
# Box and whisker plot


# ANOVA test


# Post-hoc tests
```